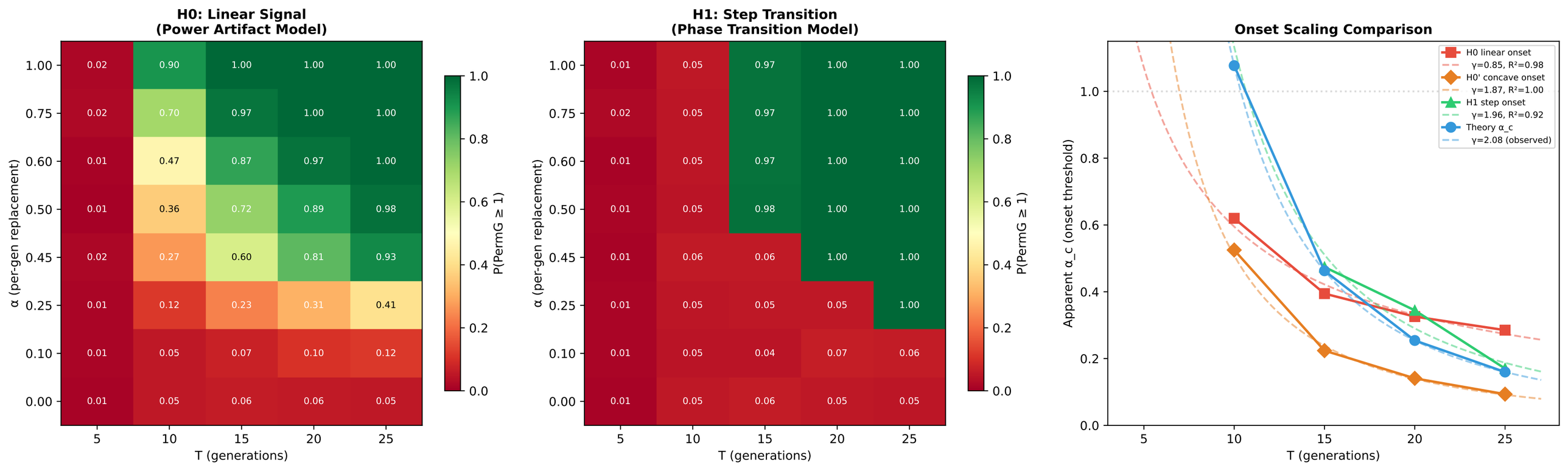
H0: Linear Signal
(Power Artifact Model)
α (per-gen replacement)
T (generations)
P(PermG ≥ 1)
1.00 0.02 0.90 1.00 1.00 1.00
0.75 0.02 0.70 0.97 1.00 1.00
0.60 0.01 0.47 0.87 0.97 1.00
0.50 0.01 0.36 0.72 0.89 0.98
0.45 0.02 0.27 0.60 0.81 0.93
0.25 0.01 0.12 0.23 0.31 0.41
0.10 0.01 0.05 0.07 0.10 0.12
0.00 0.01 0.05 0.06 0.06 0.05
5 10 15 20 25
0.0 0.2 0.4 0.6 0.8 1.0
H1: Step Transition
(Phase Transition Model)
α (per-gen replacement)
T (generations)
P(PermG ≥ 1)
1.00 0.01 0.05 0.97 1.00 1.00
0.75 0.02 0.05 0.97 1.00 1.00
0.60 0.01 0.05 0.97 1.00 1.00
0.50 0.01 0.05 0.98 1.00 1.00
0.45 0.01 0.06 0.06 1.00 1.00
0.25 0.01 0.05 0.05 0.05 1.00
0.10 0.01 0.05 0.04 0.07 0.06
0.00 0.01 0.05 0.06 0.05 0.05
5 10 15 20 25
0.0 0.2 0.4 0.6 0.8 1.0
Onset Scaling Comparison
Apparent α_c (onset threshold)
T (generations)
H0 linear onset
γ=0.85, R²=0.98
H0' concave onset
γ=1.87, R²=1.00
H1 step onset
γ=1.96, R²=0.92
Theory α_c
γ=2.08 (observed)
0.0 0.2 0.4 0.6 0.8 1.0
5 10 15 20 25